SURAJ MISHRA

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EDUCATION

Northeastern University, M.S. Computer Science (GPA: 3.85/4.00) — Boston, Massachusetts
Relevant Courses: Web Development, Program Design Paradigm, Natural Language Processing — **Sep 2023 - Dec 2025**

SRM Institute of Science and Technology, B.Tech Computer Science (GPA: 9.03/10.00) — India
Relevant Courses: Data structures and algorithms, Artificial Intelligence, Object Oriented Programming, DBMS — **Jul 2019 - Jun 2023**

SKILLS

Languages: Java, Python, C++, C#, JavaScript, TypeScript, SQL
Web Technologies: React, Next.js, Node.js, Express.js, REST APIs
Databases & Cloud: PostgreSQL, MongoDB, NoSQL, Vector Databases, AWS, CI/CD
Core Competencies: Data Structures, Algorithms, Object-Oriented Programming (OOP), Distributed Systems, Microservices, System Design, Debugging & Troubleshooting, Agile, Git
AI/LLM Tools: Claude Code, GitHub Copilot, GPT-4 API

WORK EXPERIENCE

Threadwire LLC — Boston, MA, USA
Software Engineer — **Dec 2025 - Sep 2026**

- Spearheaded **full-stack engineering** across **two production SaaS products**: a cloud-native **AI manufacturing intelligence platform** for **PLM, ERP, and MES** workflows, and **AthenaBot Slidemaker**, a GenAI-powered presentation tool built on **Next.js** and **GPT-4**
- Reduced AthenaBot Slidemaker's presentation generation time by **80% (4+ min to 50s)** via parallel slide generation using **Promise.all()** for concurrent **GPT-4 API** calls, with retry logic and rate-limit handling for fault-tolerant, resilient throughput
- Configured and deployed **RAG pipeline** using **LangChain** and **Pinecone** cloud-native vector database, integrating **OpenAI embeddings** to enable natural language queries across distributed manufacturing databases including sales orders, work orders, inventory, and risk records
- Implemented **multi-tenant SaaS architecture** with secure customer data isolation, **JWT/OAuth 2.0 authentication**, and scope-based API authorization, with **MongoDB Change Streams** enabling fault-tolerant, real-time data synchronization across the distributed platform
- Built interactive **D3.js graph visualization** for digital thread exploration, enabling manufacturing customers to trace complex relationships between products, BOMs, work orders, and blockers across the full production lifecycle
- Maintained production environment on **AWS EC2** with **PM2** process management, demonstrating operational excellence by troubleshooting production issues and monitoring API analytics across enterprise system integrations including **SAP, Oracle, and Salesforce**

IpserLab LLC — Boston, MA, USA
Software Engineer Intern — **Jan 2025 - Jun 2025**

- Developed and deployed quiz creation platform using **Claude Code** for AI-assisted development, enabling customizable assessments with multimedia support via **React, Java, and PostgreSQL**, serving **50+ active users** across **3 educational institutions**
- Accelerated API response times **30% (800ms to 560ms)** via **JDBC connection pooling** and query optimization across **200+ daily requests**, profiling slow endpoints to eliminate redundant database round-trips
- Integrated leaderboards and analytics dashboards with **Chart.js** for real-time visualization, tracking metrics across **15+ quiz sessions**
- Restructured database schema and troubleshot **12 SQL queries**, cutting execution time **25%** across **50+ concurrent sessions**
- Enhanced frontend accessibility achieving **95+ Lighthouse score** and cross-platform compatibility via **WCAG-aligned** UX best practices

Northeastern University — Boston, MA, USA
Graduate Teaching Assistant, Web Development — **Jun 2024 - Dec 2024**

- Mentored **60+ students** in full-stack development (**React, Node.js, Express.js, MongoDB, JavaScript**) through tutorials and workshops
- Conducted code reviews of **200+ assignments** evaluating debugging practices and maintainable code quality, improving average assignment scores by **15%** through **10 weekly office hours**
- Managed course forum (**200+ posts/semester**) using **Agile methodologies**, reducing unanswered queries by **40%**
- Evaluated student projects across the full **software development lifecycle (SDLC)**, providing technical feedback on code quality, **design patterns**, and **software architecture** using industry-standard practices

PROJECTS

Roommate Peacekeeper - Household Management Web Application — **Dec 2025 - Jan 2026**

- Created production-grade, fault-tolerant household management platform using **React, TypeScript, and Supabase** across **25+ reusable components** and **2,500+ lines of code**, featuring real-time chore tracking, expense splitting, and an algorithmic fairness scoring system
- Engineered real-time data sync via **Supabase PostgreSQL subscriptions**, debugging a **race condition** in authentication routing that reduced session inconsistency on page refresh from intermittent failures to **zero**
- Deployed on **Vercel** achieving **95+ Lighthouse score** across performance, accessibility, and best practices with full mobile responsiveness

Image Processing & Enhancement Application — **Sep 2024 - Dec 2024**

- Programmed **C++**-based image processing system inspired by Adobe Photoshop and GIMP, implementing **7 algorithms** (blur, sharpen, sepia, greyscale, color correction, brightness, contrast) with **MVC architecture** across **25+ classes** and **3,000+ lines of code**
- Applied **7 object-oriented design patterns** (Adapter, Facade, Builder, Command, Decorator, Strategy, Visitor) reducing cyclomatic complexity by **35%** and achieving **90% test coverage** with **Google Test**
- Optimized **multi-threaded system** supporting **100+ concurrent operations** across JPG, PNG, PPM formats, reducing average processing time by **20% (500ms to 400ms per image)**

Kanbas React Web Application — **Jan 2024 - Apr 2024**

- Crafted responsive educational platform inspired by Canvas LMS with **40+ reusable React components**, **Redux state management**, and a **GenAI-powered agent via GPT-4**, reducing initial load time from **3.2s to 2.7s** via code splitting and lazy loading
- Established scalable, maintainable component architecture with **85% code reusability** across **2,000+ lines** and **60+ files**
- Modeled **MongoDB collections** for courses, assignments, and enrollment, supporting flexible schema design across **100+ concurrent users**